Malignant Melanoma (MEL)

Case DRM-7F90425F | Source: ISIC_0034329.jpg | Analysed 2026-08-10T11:52:23 | 4-view TTA | 214 ms

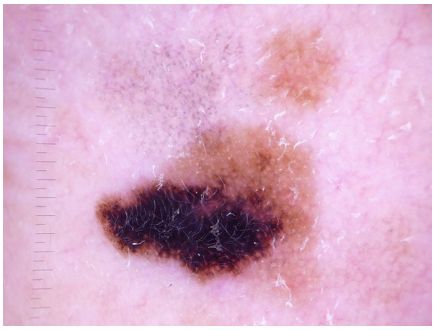
URGENT
Urgent specialist review indicated

60.8%
model confidence

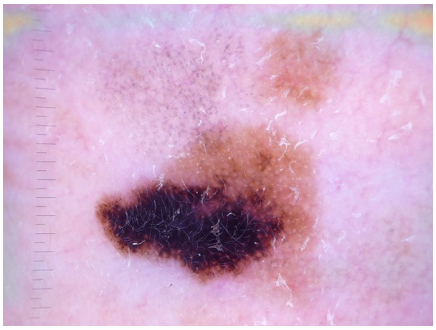
61.1%
malignant + pre-malignant

The model places substantial probability mass on a malignant class. Escalate on the urgent suspected-skin-cancer pathway.

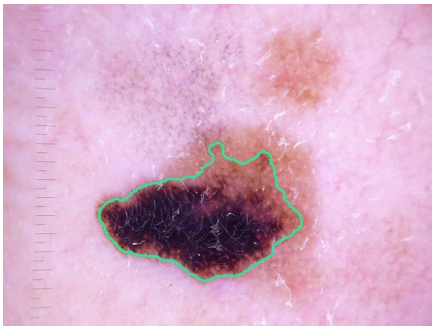
IMAGING



Submitted image



Eigen-CAM attention



Segmented lesion

Attention concentration 20% of saliency mass in the top decile. Attention is diffuse or off-centre; interpret the prediction with caution.

CLASS POSTERIOR

Class	Diagnosis	Category	P
MEL	Melanoma	Malignant	**60.8%**
NV	Melanocytic Nevus	Benign	**36.2%**
BKL	Benign Keratosis	Benign	**2.7%**
AKIEC	Actinic Keratosis	Pre-malignant	**0.2%**
DF	Dermatofibroma	Benign	**0.0%**
BCC	Basal Cell Carcinoma	Malignant	**0.0%**
VASC	Vascular Lesion	Benign	**0.0%**

Top-two margin 24.7 pts | normalised entropy 0.40 | TTA agreement 75% | peak detector score 0.912

INTERPRETATION

The leading class is Malignant Melanoma (MEL) at 60.8% posterior probability. The classifier favours this class, though the margin over the alternatives is moderate. The principal differential is Melanocytic Nevus (NV) at 36.2%. Urgent dermatology referral for dermoscopic review and excisional biopsy with narrow margins. Do not shave-biopsy a suspected melanoma: full-thickness sampling is required for accurate Breslow depth measurement, which drives staging.

Expected dermoscopic features for Melanoma:

- Atypical pigment network with irregular thickened lines
- Irregular streaks, pseudopods or radial streaming at the periphery
- Blue-white veil overlying raised areas
- Multiple colours (>3) and irregular dots or globules
- Regression structures / peppering

MORPHOMETRY - ABCD RULE OF DERMOSCOPY (STOLZ)

Axis	Measurement	Score	Weight
A - Asymmetry	16.7% non-overlap under axial reflection	2 / 2	1.3
B - Border	4 of 8 sectors with abrupt cut-off; compactness 0.66	4 / 8	0.1
C - Colour	light-brown, dark-brown, black	3 / 6	0.5
D - Structures	5 distinct colour-texture regions (approximation)	5 / 5	0.5

Total Dermoscopy Score 7.0 - Above 5.45 - morphology is highly suspicious for melanoma.

Lesion diameter 284 px (47% of frame width). Absolute size in millimetres requires a calibrated scale in frame. The E axis (evolution) cannot be assessed from a single image. Serial photography at 3-6 month intervals is required.

SUGGESTED MANAGEMENT

Urgent dermatology referral for dermoscopic review and excisional biopsy with narrow margins. Do not shave-biopsy a suspected melanoma: full-thickness sampling is required for accurate Breslow depth measurement, which drives staging.

Urgency: Refer within 2 weeks (or per local urgent suspected-cancer pathway).

IMAGE QUALITY

Quality index 91/100 | focus 142 | resolution 600x450 | No quality issues detected.

IMPORTANT. Dermora AI is an academic decision-support prototype. The underlying model was trained on the ISIC 2018 Task 3 (HAM10000) dataset, which over-represents fair skin types and does not cover every lesion class encountered in practice. Output is a probability distribution, not a diagnosis, and carries no regulatory clearance for clinical use. It must not be used to make or defer treatment decisions. Any lesion of concern requires in-person assessment by a qualified clinician, and histopathology remains the diagnostic standard.